
ENGINEERING TRIPOS PART IIA

ELECTRICAL AND INFORMATION ENGINEERING TEACHING LABORATORY

MODULE EXPERIMENT 3F1

FLIGHT CONTROL

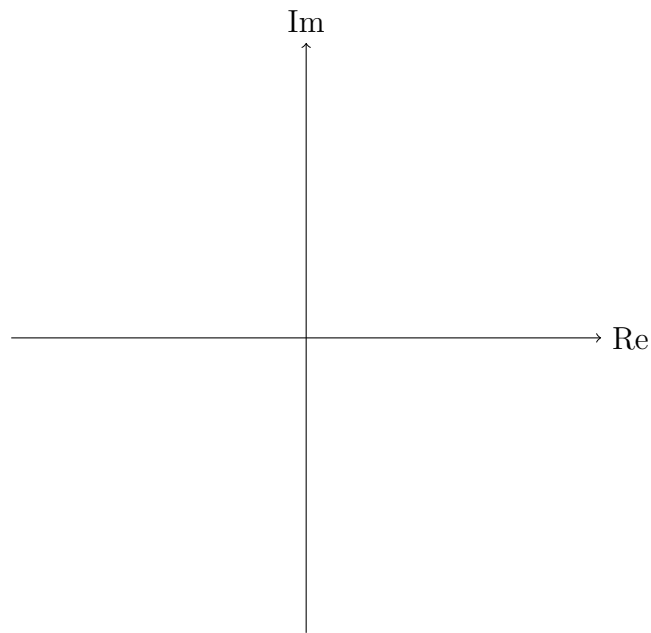
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Objectives:

- Simulation of various aircraft models on the computer.
- Study real-time (manual) control and the limitations imposed by time delays.
- Design of a simple autopilot.
- Illustrate frequency response concepts in analogue and digital control systems, conditions for oscillation in feedback systems and stability.
- Gain familiarity with Python and Jupyter Notebooks.

3F1 Flight Control Lab

1. (§2.2 Modelling Manual Control) Nyquist diagram (from Bode diagram) for controller in series with plant.



Nyquist diagram

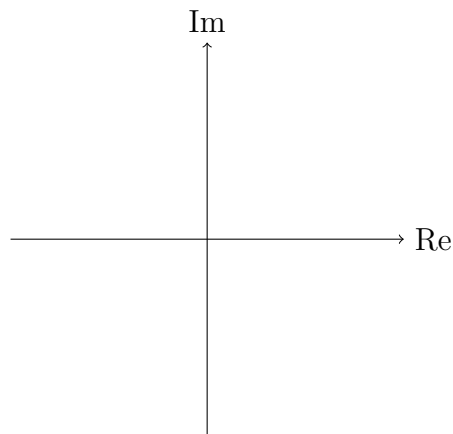
2. (§2.2 Modelling Manual Control) Are you using any integral action? Give a brief explanation. What does this imply about the accuracy of the model of the human controller?

3. (§2.3 Pilot Induced Oscillation) Explain the oscillation of the feedback loop. How does your observed period of oscillation compare to the theoretical prediction?

4. (§2.3 Pilot Induced Oscillation) Can you give a rough guideline to the control designer to make PIO less likely?

5. (§2.4 Sinusoidal Disturbances) Was your manual input able to reduce the error (as compared to providing no input)?

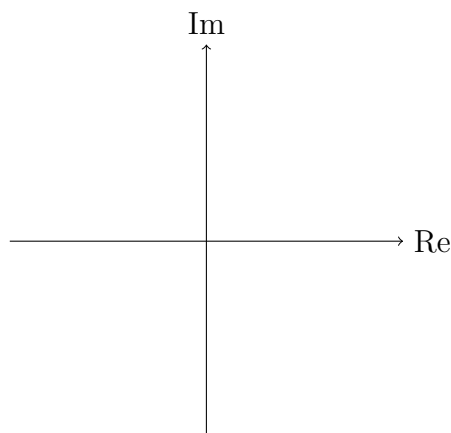
6. (§2.5 An Unstable Aircraft) Nyquist diagram for $G_2(s)$.



Nyquist diagram of $G_2(s)$

7. (§2.5 An Unstable Aircraft) Explain, using the Nyquist criterion, why the feed-back system is stable with a proportional gain greater than 0.5.

8. (§2.5 An Unstable Aircraft) Sketch of a Nyquist diagram for $G_2(s)$. with a small time delay D .



Nyquist diagram with small time delay

9. (§3.4 Integrator Wind-up) Explain how you calculated the bound on Q .